PROJECT TOPIC: CodeONE

Consolidate Your Competitive Programming Journey

Group No.: 189

Project Group Members:

1. Ayush Raj (4F/2315000557)
2. Astha Goswami (4E/2315000506)
3. Divyanshi Kulshrestha (4G/2315000790)
4. Anvesha Singh (4G/2315000401)
5. Ishendra Rai (4G/2315000994)

Project Supervisor: Mr. Raju Singh, Assistant Professor

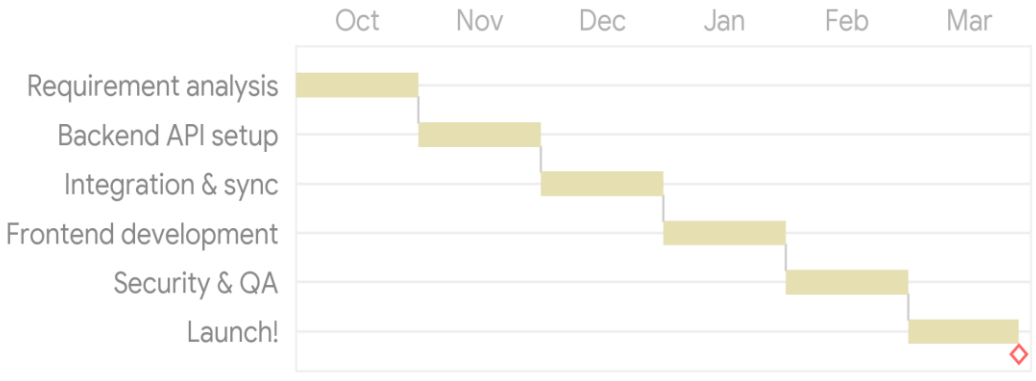
About the Project: CodeONE is a security-hardened, full-stack competitive programming analytics platform designed to serve as a unified performance intelligence layer for developers, students, and interview aspirants. Rather than relying on fragile or unauthorized web scraping, it employs an isolated Platform Adapter Layer that integrates permitted coding sources—prioritizing the official Codeforces API first, followed by AtCoder (via AtCoder Problems), Codewars, CLIST.by for contest discovery, and a manual data import fallback for platforms whose Terms of Service restrict automated scraping (such as LeetCode and CodeChef). Built on an API-first modular monolith architecture, the backend utilizes Node.js (v22+), Express, PostgreSQL (v15+), and Prisma ORM with staged Redis caching, while the frontend delivers a responsive React 18 and Vite workspace featuring 365-day activity heatmaps, global streak tracking, problem bookmarking and personal notes, and rule-based topic mastery insights. To protect user data and prevent API abuse, CodeONE enforces Argon2id/bcrypt hashing, verified Google/GitHub OAuth 2.0, 15-minute JWT access tokens with 7-day rotating refresh tokens and reuse detection, four-role default-deny RBAC, strict IDOR protection, and append-only audit logging.

Motivation: Competitive programmers frequently face fragmented progress tracking, streak blindness, pattern ignorance, and contest chaos when switching between five or more isolated platform dashboards. Solving this requires rigorous engineering: platforms expose heterogeneous data models, enforce strict rate limits, and prohibit unauthorized crawling or credential harvesting. Furthermore, multi-platform aggregators are prime targets for broken object-level authorization (IDOR), credential stuffing, token replay, and OWASP Top 10 vulnerabilities. CodeONE serves as an ideal Final Year major project because it combines full-stack systems architecture—relational schema design separating mutable problem state from immutable submission histories, adapter-based API normalization, and staged caching—with a verifiable cybersecurity and API-compliance framework that empowers students to systematically overcome technical weaknesses and improve employability.

Innovation:

- **Unified Global Streak & Submission Engine:** Separates current problem state (user_problems) from timestamped submission records (submissions) to compute timezone-aware global streaks and a 365-day contribution heatmap across linked accounts, with an optional clearly labelled streak recovery mode.
- **Policy-Compliant Platform Adapter Architecture:** Isolates external integrations behind a unified PlatformAdapter interface supporting Codeforces (official API), AtCoder Problems, Codewars, CLIST.by contest aggregation, and manual data imports, while surfacing explicit per-platform states (connected, partial data, rate limited, source unavailable).
- **Zero-Credential Handle Ownership Verification:** Eliminates third-party password and session-cookie storage risks by verifying public platform handles via one-time profile challenge codes or official OAuth, paired with verified-email Google and GitHub OAuth 2.0 authentication.
- **Hybrid Token Lifecycle & Reuse Detection:** Implements short-lived 15-minute in-memory JWT access tokens and 7-day HttpOnly, Secure, SameSite=Strict refresh cookies backed by server-side SHA-256 token hashing (user_sessions, refresh_tokens) and automatic session-family revocation upon token reuse detection.
- **Defense-in-Depth Security & Rule-Based Weakness Detection:** Enforces four-role default-deny RBAC (User, Moderator, Admin, Auditor), user_id-scoped IDOR prevention, Joi schema validation, parameterized Prisma queries, append-only audit_logs, and explainable Level 1 rule-based topic weakness detection across 20+ algorithmic categories.

Project Planning: The project development phases will be carried out between October 2026 and March 2027. Below is the Gantt chart illustrating our planned project activities over this duration.



- **Oct 2026:** Requirement analysis, API-first modular monolith architecture design, external platform API terms/compliance audit, and 15-table PostgreSQL/Prisma schema design (users, platform_accounts, submissions, user_problems, user_sessions, audit_logs).

- **Nov 2026:** Backend API setup (Node.js 22+ / Express), authentication module (Argon2id/bcrypt, Google & GitHub OAuth 2.0), 15-minute JWT + 7-day rotating refresh token lifecycle with reuse detection, and 4-role RBAC middleware.

- **Dec 2026:** PlatformAdapter interface implementation, Codeforces official connector, public-handle ownership verification, asynchronous sync_logs tracking, rate limiting, and CLIST.by contest ingestion scheduler.

- **Jan 2027:** Frontend development (React 18, Vite, Tailwind CSS, TanStack Query, Recharts), unified analytics dashboard, 365-day streak heatmap, problem management (status, notes, bookmarks), and additional adapters (AtCoder Problems, Codewars, manual import).

- **Feb 2027:** Analytics engine & Level 1 rule-based topic weakness detection, staged Redis caching for heavy aggregates, append-only security audit logging, and automated security/IDOR/token-reuse testing via Jest and Supertest.

- **Mar 2027:** Docker containerization (docker-compose), GitHub Actions CI/CD quality gates, Nginx reverse proxy & Let's Encrypt TLS hardening, documentation verification, and final deployment.

Tools required:

Hardware Requirements: A standard PC/Laptop (minimum 8GB RAM, Intel Core i5 processor or equivalent), and a stable Internet connection for external API communication.

Software & Security Requirements: Node.js (v22+ LTS), Express.js, React.js (v18+), Vite, Tailwind CSS, TanStack Query, Recharts, PostgreSQL (v15+), Prisma ORM, Redis (v7+, optional caching), Argon2id / bcrypt, JSON Web Tokens (jsonwebtoken), Passport.js (OAuth 2.0), Joi Validator, Helmet, express-rate-limit, Jest, Supertest, Docker, GitHub Actions, Nginx, Let's Encrypt, and Git.

Signature of Project Supervisor: _____